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ABSTRACT

Context. Optional, leave empty if necessary. The heading "Context" is used when needed to give background information on the research conducted in the paper
Aims. Mandatory. The objectives of the paper are defined here.
Methods. Mandatory. The methods of the investigation are outlined here
Results. Mandatory. The results are summarized here.
Conclusions. Optional, leave empty if necessary. "Conclusions" can be used to explicit the general conclusions that can be drawn from the paper.

Key words. gravitational waves – strong gravitational lensing – parameter estimation – sky localization – binary black holes – LIGO

1. Introduction

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2. Results: sky localization with H1+L1

We now assess the improvement in sky localization achieved by the coherent joint analysis of strongly lensed image pairs, relative to the analysis of each image in isolation, for the two-detector H1+L1 (HL) network under projected O5a sensitivity (LIGO Scientific Collaboration 2025), across the full sample of 157 analyzed events.

2.1. 90% credible sky area

For each of the 157 events, we compute the 90% credible sky area A_{90} (Eq. (**??**)) from the single-image posteriors of image 1 and image 2, and from the joint posterior. We define the best single-image area as $A_{90}^{\rm best} \equiv \min(A_{90}^{(1)}, A_{90}^{(2)})$, and the per-event improvement factor from combining the two images as

$$\mathrm{IF}_{\rm joint} \equiv A_{90}^{\rm best}/A_{90}^{\rm joint}. \quad (1)$$

Figure 1 shows the cumulative and probability density distributions of A_{90} for the best single-image and joint analyses. The joint distribution is shifted to smaller areas across the entire sample: the median 90% credible area drops from 562.5 deg^2 (best single image) to 90.7 deg^2 (joint), a median improvement factor of $\mathrm{IF}_{\rm joint} = 5.19$. The joint analysis reduces the sky area of every one of the 157 events, with $\mathrm{IF}_{\rm joint}$ between 1.11 and 27.90. The improvement is therefore general, and does not come from just a few events with a favorable sky position. Six events (3.8 %) reach a joint area below 10 deg^2, the benchmark commonly adopted for host-galaxy identification and multimessenger follow-up with wide-field survey telescopes (Abbott et al. 2020).

2.2. Dependence on network SNR

Figure 2 shows A_{90} as a function of the per-image network SNR, for each image and for the joint analysis. The single-image areas show substantial scatter at fixed SNR, reflecting the dominant role of sky position and antenna pattern orientation, rather than SNR alone, in setting the localization precision of a two-detector network (Sect. **??**). The joint analysis consistently reduces A_{90} relative to either single image, with the largest gains occurring for events whose single-image areas are largest to begin with.

2.3. Constellation localization

We further characterize the localization using the fraction of posterior probability contained in the dominant IAU constellation (Singer & Price 2016). For single-image analyses the median fraction in the top constellation is 41.75 % (best of the two images). The joint analysis raises the median to 69.3 %, with 121/157 events (77.1 %) above 50 % and 49/157 events (31.2 %) above 90 %.

Comparing against the published sky maps of 104 GWTC-5 events (The LIGO Scientific Collaboration et al. 2026), we find that the joint lensed-pair analysis with only two detectors reaches

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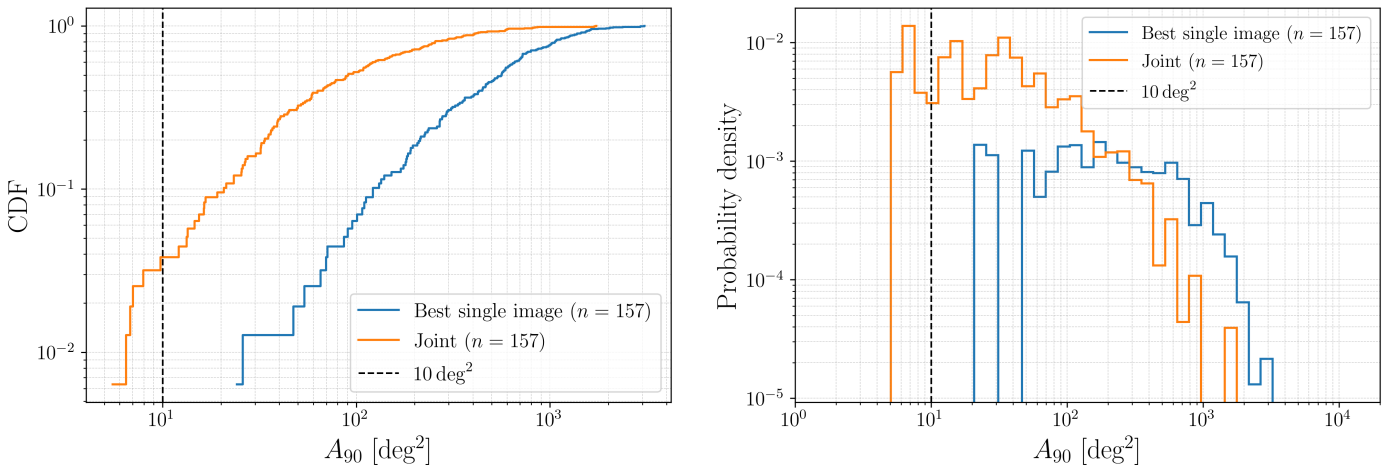


Fig. 1: Cumulative (left) and probability density (right) distributions of the 90% credible sky area, for the best single-image analysis and the joint analysis, across the 157 analyzed events. The vertical dashed line marks 10 deg^2.

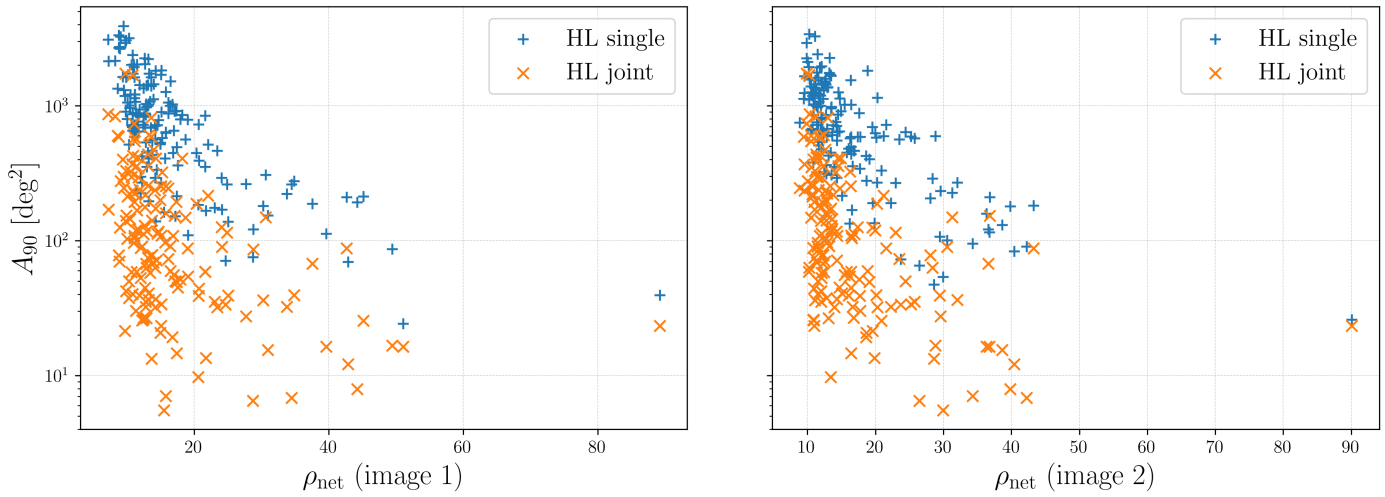


Fig. 2: 90% credible sky area versus network SNR for image 1 (left) and image 2 (right). Blue crosses: single-image analysis; orange crosses: joint analysis for the same event.

constellation fractions comparable to those achieved by the three-detector HLV network on real data, whereas real HL-only events are confined below $\sim 50\,\%$ (Fig. 3). This places the improvement from the joint lensed analysis on a comparable footing to the localization gain from adding a third detector, using only two detectors and a repeated observation of the same source, a connection we examine directly in Sect. 3.

2.4. Comparison with previous work

Hannuksela et al. (2020) demonstrated that quadruply lensed events observed by the full LIGO/Virgo/KAGRA network can, combined with electromagnetic constraints on the lens galaxy, be localized to one or a few candidate host galaxies. Li & Hannuksela (2026) performed the first systematic study of repeated-signal localization using Bayestar (Singer & Price 2016) sky maps on a simulated population observed with the three-detector HLV network, finding roughly an order-of-magnitude reduction in the 90% credible area from combining two images. The smaller median improvement factor found here (5.19) is consistent with the more conservative two-detector con-

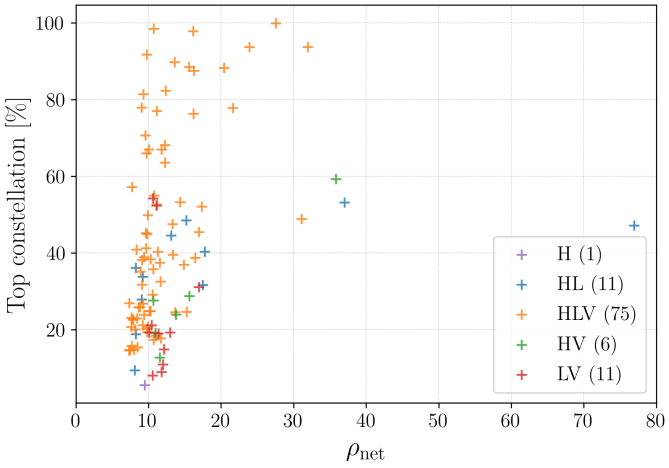


Fig. 3: Probability fraction in the dominant constellation versus network SNR for 104 GWTC-5 events, colored by detector network. The HL subset, matching the detector configuration used in this work, is confined below $\sim 50\,\%$.

figuration adopted in this work: with only H1 and L1, single-image localization already relies more heavily on antenna pattern information than on triangulation, leaving comparatively less room for a second image, which acts through the same antenna-pattern channel, to improve upon it. Both studies nonetheless converge on the same qualitative conclusion, namely that combining two lensed images yields a substantial and consistent sky localization gain regardless of detector network or analysis method. Section 3 tests this conclusion directly by asking how the gain from the lensed pair interacts with growing the detector network itself.

3. Adding a third detector: Virgo

The results of Sect. 2 isolate the gain from a coherent joint analysis of two lensed images within a fixed two-detector network. We now ask a complementary question: how much further does sky localization improve when a third detector, Virgo, is added to the network, and how does that gain depend on the geometry of the event relative to the H1-L1 baseline and on its signal-to-noise ratio?

3.1. A stratified subsample, not a random one

Re-running the full 157-event population with Virgo added is computationally prohibitive within the scope of this work. We instead select a stratified subsample of 40 events, designed to probe five distinct regimes of expected Virgo benefit, rather than drawing a random subset: `core` (the 16 events with the largest fractional SNR gain from adding V1, computed with the exact inner-product SNR estimator of GWSNR; Phurailatpam & Hannuksela 2026), `geom` (the 10 events for which neither lensed image is well aligned with the H1-L1 baseline, quantified by the angle between the source direction and the baseline at each image's arrival time, so that triangulation with H1+L1 alone is least reliable), `control` (5 events with low expected SNR gain and good H1-L1 alignment, included as a null case), `threshold` (5 events whose H1+L1 joint area sits close to the $10\,\mathrm{deg}^2$ benchmark of Sect. 2.1, where a modest absolute change in area has an outsized practical effect on follow-up feasibility), and `low-SNR` (4 events with H1+L1 network SNR $\rho_{\mathrm{HL}} < 15$, among the faintest in the population, included to test how much Virgo can help when the signal itself is weak).

Figure 4 shows the network SNR of every event with and without Virgo, with the five groups highlighted. We checked that this selection does not leave out events likely to benefit strongly from Virgo. Every event in the population with a fractional SNR gain above $\sim 19\,\%$ belongs to `core`, and all unselected events with $A_{90}^{\mathrm{joint,\,HL}} > 200\,\mathrm{deg}^2$ have a Virgo SNR $\rho_{\mathrm{V1}} = (\rho_{\mathrm{HLV}}^2 - \rho_{\mathrm{HL}}^2)^{1/2} \lesssim 10$, below that of the `core` events with similar HL areas.

Because this selection deliberately overrepresents events favorable to Virgo, the median statistics reported below describe this stratified subsample and *must not* be compared directly to the population-level medians of Sect. 2. Instead, every comparison in this section is made event by event: the same 40 events are re-analyzed with the H1+L1+V1 (HLV) network, using the Virgo O5 low-sensitivity design curve (LIGO Scientific, Virgo and KAGRA Collaborations 2022), under the same pipeline, priors, and waveform model as the H1+L1 (HL) analysis, and the two outcomes for each event are compared directly. The full per-event table is given in Appendix A.

3.2. Sky area: HL versus HLV

Figure 5 shows the joint 90% credible area with HLV against the same quantity with HL, for all 40 events. Every event falls below the diagonal: adding Virgo improves the localization of every event in the subsample, with no exceptions. The per-event improvement factor, $\mathrm{IF}_{\mathrm{Virgo}} \equiv A_{90}^{\mathrm{joint,\,HL}}/A_{90}^{\mathrm{joint,\,HLV}}$, has a median of 11.5 across the subsample (range 2.4–93.8), and the median joint area drops from $109.0\,\mathrm{deg}^2$ (HL) to $10.0\,\mathrm{deg}^2$ (HLV). Twenty of the 40 events (50 %) fall below the $10\,\mathrm{deg}^2$ benchmark with Virgo, compared to a single event with H1+L1 alone. The median dominant-constellation fraction rises correspondingly, from 65.6 % (HL) to 99.3 % (HLV), and the number of events with more than 90 % of the probability in a single constellation grows from 13 to 24. The constellation fraction is a coarse measure, however: for 5 of the 40 events it decreases even though the area shrinks, since a smaller region can still straddle a constellation boundary.

Figure 6 shows the same comparison per event, as a slope chart grouped by selection category, and Fig. 7 summarizes the resulting improvement factors by group (see also Table 1). The gain is clearly heterogeneous across groups: the `core` and `geom` groups, selected respectively for high expected SNR gain and for poor H1-L1 geometric alignment, show the largest improvements (median $\mathrm{IF}_{\mathrm{Virgo}} = 14.6$ and 21.8), while the `control` group, selected as a null case of low SNR gain and favorable H1-L1 alignment, shows the smallest (median 3.4). The `threshold` group, selected independently of either criterion, falls in between (median 9.2). This ordering confirms that the selection criteria used to build the subsample track the physical drivers of the Virgo gain: events that are geometrically degenerate under H1+L1, or that happen to be loud in Virgo, benefit the most, while events already well constrained by H1+L1 triangulation gain comparatively little from a third detector. The `low-SNR` group (median 7.7) does not fit this simple ordering; we discuss it in Sect. 3.3.

Table 1: Joint 90% credible area and Virgo improvement factor by selection group.

Group	N	$\tilde{A}_{90}^{\mathrm{HL}}$ [deg^2]	$\tilde{A}_{90}^{\mathrm{HLV}}$ [deg^2]	$\widetilde{\mathrm{IF}}_{\mathrm{Virgo}}$	$N_{<10}^{\mathrm{HL}}$	$N_{<10}^{\mathrm{HLV}}$
Core	16	186.6	12.0	14.6	1	6
Geom	10	335.9	5.9	21.8	0	7
Control	5	36.3	10.6	3.4	0	2
Threshold	5	21.3	2.0	9.2	0	5
Low SNR	4	271.2	54.6	7.7	0	0
All	40	109.0	10.0	11.5	1	20

Notes. A tilde denotes the group median. $\tilde{A}_{90}$ is the median joint 90% credible area with each network, and $N_{<10}$ the number of events with a joint area below $10\,\mathrm{deg}^2$.

3.3. Low-SNR events: a sizeable factor, but areas still too large

The `low-SNR` group was included as a second null case: with $\rho_{\mathrm{HL}} < 15$, these events carry little signal in Virgo, and we expected the gain from adding V1 to be small. Its median improvement factor, $\mathrm{IF}_{\mathrm{Virgo}} = 7.7$, is nevertheless larger than that of the `control` group (3.4). The reason is that $\mathrm{IF}_{\mathrm{Virgo}}$ is a ratio, and it grows with the HL area it starts from. The `low-SNR` events start from large HL areas (170–601 deg^2, median 271 deg^2), while

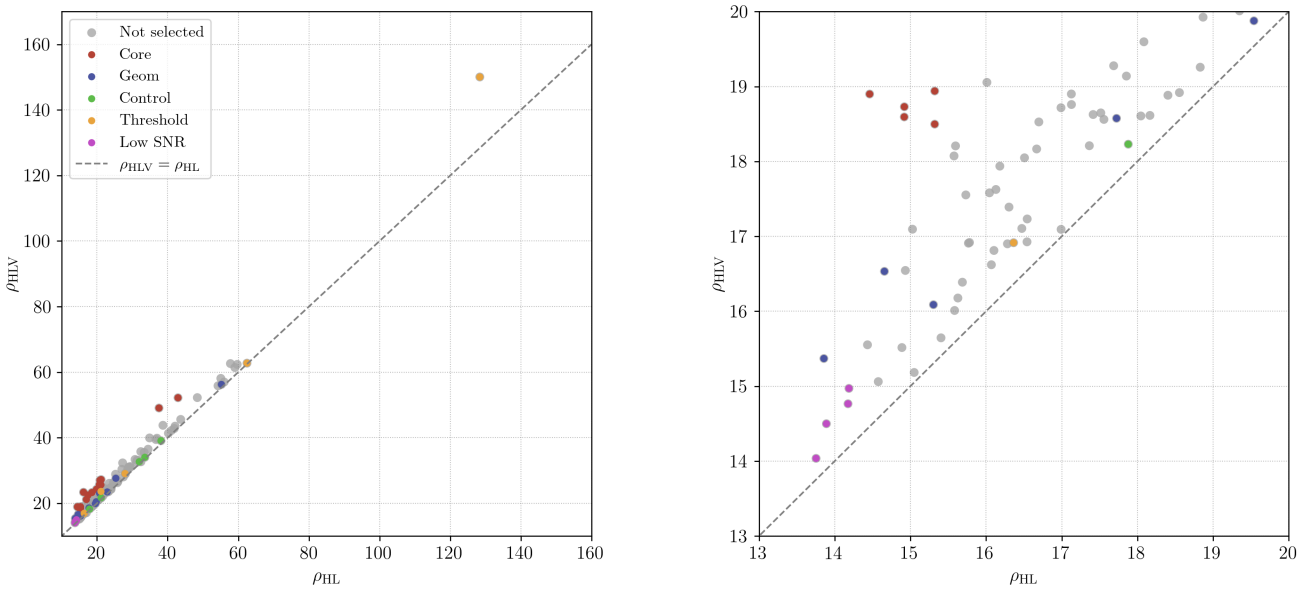


Fig. 4: Network SNR with H1+L1+V1 (ρ_{HLV}) versus H1+L1 (ρ_{HL}) for the 157 events. Grey points are events not selected for the Virgo analysis; colored points are the 40 events of the stratified subsample, by selection group. The right panel zooms into the low-SNR end, $13 < \rho_{HL} < 20$. The dashed line marks $\rho_{HLV} = \rho_{HL}$.

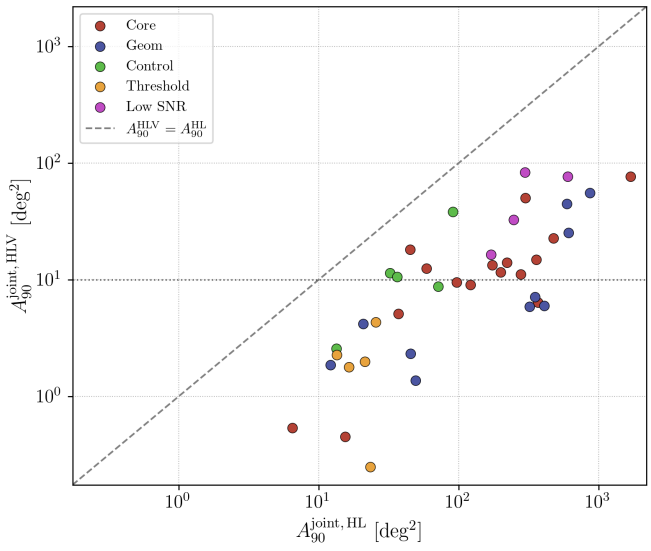


Fig. 5: Joint 90% credible area with H1+L1+V1 (HLV) versus H1+L1 (HL), for the 40-event stratified subsample, colored by selection group. The dashed line marks equal area; every point falls below it. The dotted line marks 10 deg^2.

the `control` events start from small ones (13–91 deg^2, median 36 deg^2), which leave much less room for improvement. The two groups therefore cannot be compared directly.

Comparing events at similar HL area instead reveals the expected effect of a low SNR (Fig. 5). The 12 `core` and `geom` events whose HL areas fall in the same range as the `low-SNR` group (170–601 deg^2) have a median $IF_{Virgo} = 22.5$ (range 6.0–68.6), and all four `low-SNR` events (3.6–10.4) lie below 11 of these 12. Since the `low-SNR` group contains only four events, and the comparison range is set by those same events, we read this as a clear trend rather than a precise measurement.

For follow-up, what matters is the final area rather than the factor by which it shrinks. None of the four `low-SNR` events reaches the 10 deg^2 benchmark with HLV: their joint areas remain between 16 and 83 deg^2 (median 55 deg^2), whereas half of the subsample, including all five `threshold` events and 7 of the 10 `geom` events, falls below it (Table 1). For faint lensed pairs, adding Virgo still shrinks the sky area by a sizeable factor, but less than for louder events with similar HL areas, and the final area remains too large for efficient follow-up with wide-field telescopes.

3.4. The diminishing, but still positive, value of the lensed pair

A natural question is whether the gain from combining two lensed images, the central result of Sect. 2, persists once a third detector is already available. Within the HL network, the median value of IF_{joint} (Eq. (1)) for this 40-event subsample is 4.69, consistent with the 5.19 found for the full 157-event population (Sect. 2.1) and confirming that the subsample is not biased with respect to this particular metric. Within the HLV network, the same quantity, which we denote IF_{joint}^{HLV} to distinguish it from its HL counterpart IF_{joint}^{HL}, drops to 2.18.

Figure 8 shows this effect event by event. Each point places the single-image areas of image 1 and image 2 against each other, and its color gives the ratio of the joint area to the best single-image area, $A_{90}^{joint}/A_{90}^{best} = 1/IF_{joint}$. With HL, the joint analysis often reduces the area to 0.1–0.3 times that of the best image. With HLV, the single-image areas are already smaller, typically by an order of magnitude or more, and the ratio lies mostly between 0.3 and 1: Virgo and the second image largely supply the same information.

This does not mean that combining the two lensed images becomes uninformative once Virgo is added. All but one event in the subsample still benefit from the joint analysis within the HLV network itself (Appendix A). The exception, inj122, has both single-image HLV areas below 1 deg^2, and its joint

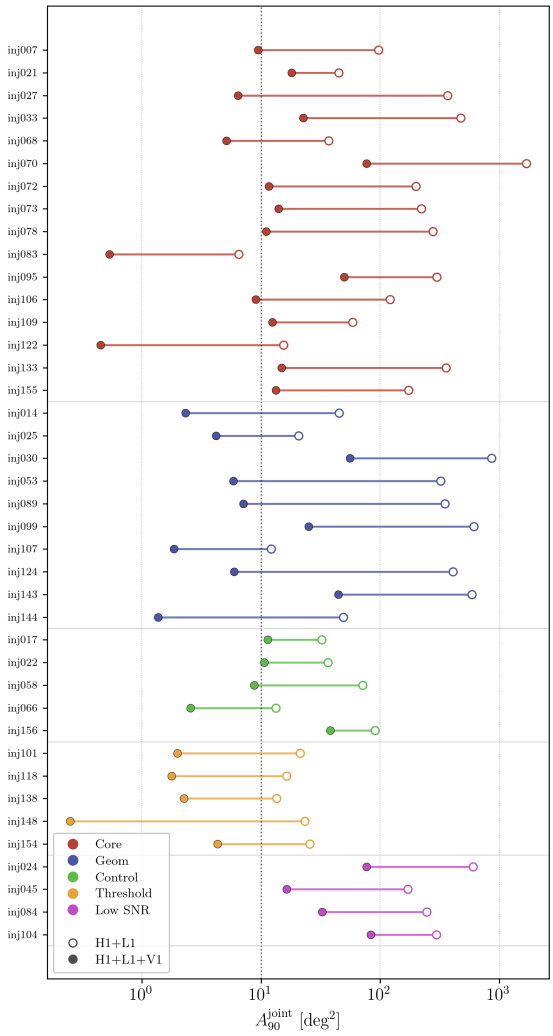


Fig. 6: Joint 90% credible area for each of the 40 events, with H1+L1 (open markers) and H1+L1+V1 (filled markers), grouped by selection category. The dotted line marks 10 deg^2.

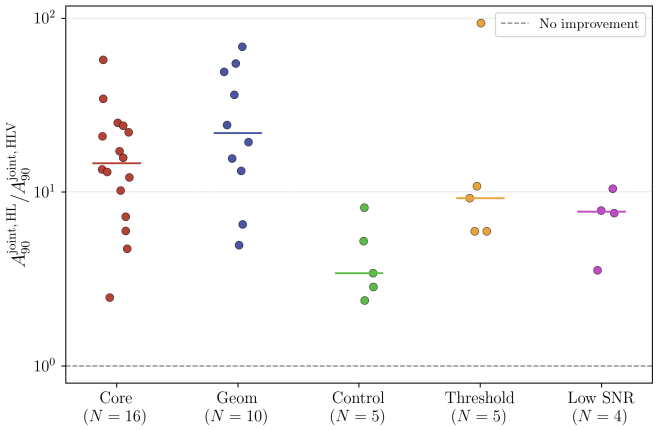


Fig. 7: Improvement factor from adding Virgo, $\mathrm{IF}_{\mathrm{Virgo}} = A_{90}^{\mathrm{joint,\,HL}}/A_{90}^{\mathrm{joint,\,HLV}}$, for each event of the Virgo subsample, grouped by selection category. Horizontal bars mark the group medians; the dashed line marks no improvement.

area (0.4 deg^2) is slightly larger than the best single-image area ($\mathrm{IF}_{\mathrm{joint}}^{\mathrm{HLV}} = 0.68$), a difference we attribute to the finite number of posterior samples and the sky-map resolution at this scale. The absolute localization achieved by the HLV joint analysis remains, by a wide margin, the best result available for any of these events. What the drop in $\mathrm{IF}_{\mathrm{joint}}$ indicates is that the specific role played by the second lensed image, namely supplying an independent triangulation baseline through its different arrival time and antenna-pattern orientation, is partially subsumed once a third detector already provides that information directly. The lensed-pair technique and a richer detector network are therefore complementary rather than redundant: the former matters most precisely in the regime probed by the `geom` group, where the detector network alone is geometrically degenerate, and its marginal contribution shrinks, without vanishing, as the network's own triangulation power increases.

3.5. Comparison with the literature

These results extend the central finding of Sect. 2.4 along a second, complementary axis: while Sect. 2 shows that repeating an observation through a lensed image pair substantially improves localization within a fixed detector network, the present section shows that growing the detector network itself delivers a comparable, and in the most favorable cases much larger, improvement, and that the two strategies interact in the physically expected way, each compensating most for the specific geometric limitations the other leaves unaddressed.

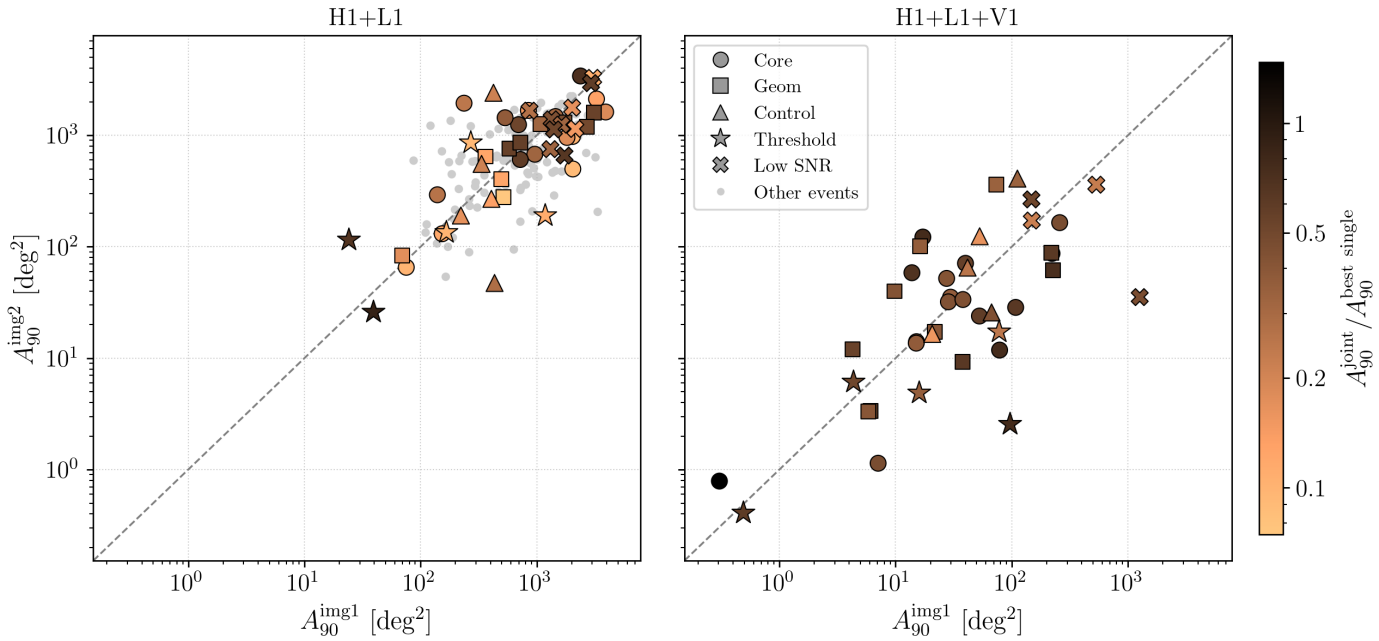


Fig. 8: 90% credible area of image 2 versus image 1, with H1+L1 (left) and H1+L1+V1 (right), for the 40 events of the Virgo subsample. Marker shape indicates the selection group, and color the ratio of the joint area to the best single-image area (darker means a smaller gain from the joint analysis). Grey dots in the left panel show the remaining 117 events of the full population, analyzed with H1+L1 only. The dashed line marks equal areas.

4. Conclusions

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Appendix A: Full per-event results for the Virgo subsample

Table A.1 lists, for each of the 40 events analyzed with H1+L1+V1 (Sect. 3), the selection group, the joint 90% credible area with H1+L1 (HL) and H1+L1+V1 (HLV), the per-event improvement factor from adding Virgo to the joint analysis ($\mathrm{IF}_{\mathrm{Virgo}}$), and the improvement factor from combining the two lensed images within each network separately ($\mathrm{IF}^{\mathrm{HL}}_{\mathrm{joint}}$, $\mathrm{IF}^{\mathrm{HLV}}_{\mathrm{joint}}$), as discussed in Sect. 3.4.

Table A.1: Per-event sky-area results for the 40-event Virgo subsample. Events are grouped by selection group and, within each group, sorted by decreasing $\mathrm{IF}_{\mathrm{Virgo}}$.

Event	Group	$A^{\mathrm{joint,\,HL}}_{90}$ [deg^2]	$A^{\mathrm{joint,\,HLV}}_{90}$ [deg^2]	$\mathrm{IF}_{\mathrm{Virgo}}$	$\mathrm{IF}^{\mathrm{HL}}_{\mathrm{joint}}$	$\mathrm{IF}^{\mathrm{HLV}}_{\mathrm{joint}}$
inj027	Core	367.8	6.4	57.64	1.89	2.20
inj122	Core	15.4	0.4	34.36	8.48	0.68
inj078	Core	276.9	11.1	25.02	7.66	2.69
inj133	Core	357.3	14.8	24.07	1.70	2.26
inj070	Core	1690.2	76.7	22.03	1.41	2.15
inj033	Core	473.0	22.5	20.98	3.06	1.78
inj072	Core	199.4	11.6	17.19	2.68	2.37
inj073	Core	221.3	14.0	15.76	3.06	1.70
inj106	Core	121.4	9.0	13.48	6.98	1.31
inj155	Core	173.7	13.3	13.03	5.51	1.29
inj083	Core	6.5	0.5	12.14	10.09	2.14
inj007	Core	96.6	9.4	10.22	10.12	1.47
inj068	Core	36.9	5.1	7.23	3.78	2.67
inj095	Core	299.1	50.0	5.98	5.41	1.74
inj109	Core	58.8	12.4	4.72	4.03	2.29
inj021	Core	44.7	18.1	2.47	11.15	1.58
inj124	Geom	407.9	5.9	68.61	1.77	2.74
inj053	Geom	321.2	5.9	54.76	1.80	1.58
inj089	Geom	350.5	7.1	49.23	3.07	2.43
inj144	Geom	49.2	1.4	36.21	8.18	2.45
inj099	Geom	609.7	25.1	24.28	2.14	2.93
inj014	Geom	45.1	2.3	19.42	8.05	1.83
inj030	Geom	865.1	55.5	15.58	1.85	1.59
inj143	Geom	590.9	44.5	13.27	2.02	1.38
inj107	Geom	12.1	1.9	6.53	5.76	1.81
inj025	Geom	20.7	4.2	4.96	13.42	2.35
inj058	Control	71.2	8.8	8.13	4.71	6.05
inj066	Control	13.3	2.6	5.22	3.55	6.47
inj022	Control	36.3	10.6	3.42	7.42	3.93
inj017	Control	32.3	11.4	2.84	5.91	2.26
inj156	Control	90.7	38.2	2.38	4.68	2.93
inj148	Threshold	23.3	0.2	93.77	1.11	1.65
inj101	Threshold	21.3	2.0	10.79	8.92	1.29
inj118	Threshold	16.4	1.8	9.24	1.48	2.75
inj138	Threshold	13.5	2.3	5.96	10.00	1.93
inj154	Threshold	25.6	4.3	5.95	10.58	4.01
inj045	Low SNR	170.4	16.3	10.44	6.70	2.17
inj024	Low SNR	600.9	76.6	7.84	2.23	1.93
inj084	Low SNR	246.4	32.5	7.58	3.05	4.58
inj104	Low SNR	296.0	83.2	3.56	5.99	4.34

Appendix B: Wide tables and figures after an appendix title: recommended method

In the PDF output, floats should be placed under their own appendix, not before the title, nor after the title of the next appendix. In short appendices, one-column floats {figure*} or {table*} will generate a blank page. To prevent this behaviour, we recommend to switch to \onecolumn and set the [ht!] parameter in your floats: please check the LaTeXcode of this appendix.

In case you have a lot of floating objects for little text and the LaTeXengine moves the floats away from their context, the command \FloatBarrier of the "placeins" package will empty the float buffer and place all stored floats in the continuity. If you still encounter problems with wide floats placement, just use the \onecolumn environment throughout the appendices.

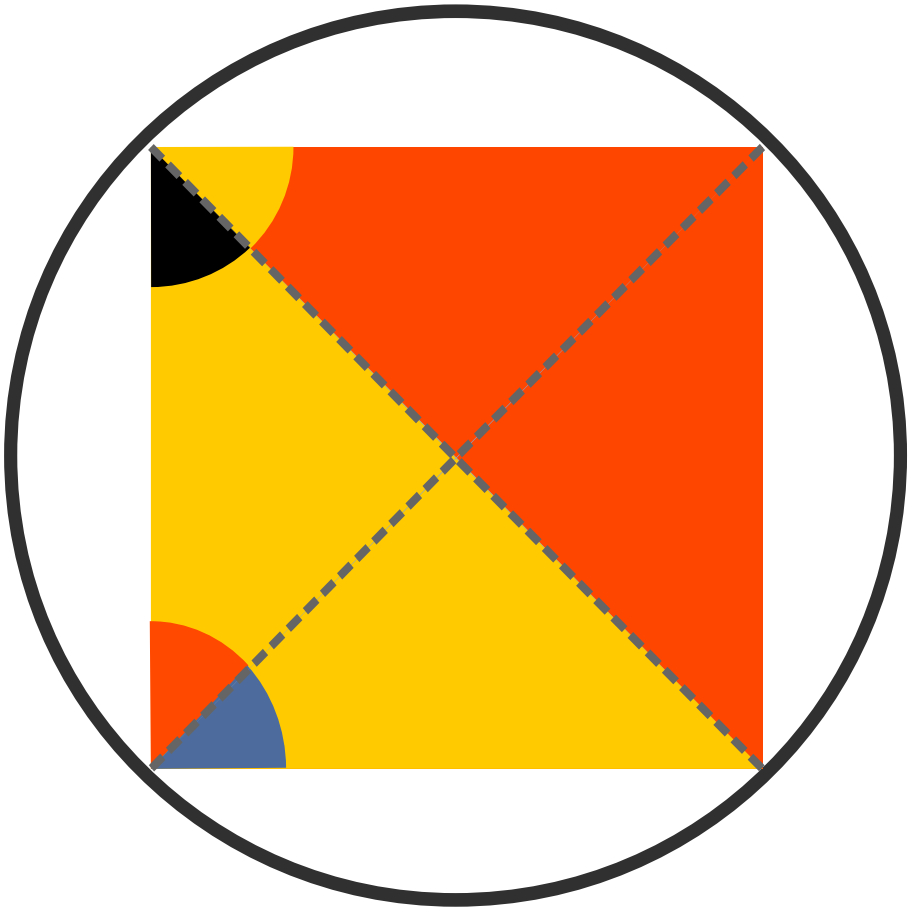


Fig. B.1: A one-column {figure*}[ht!] after a section title. If text follows like below, it is easier to finish the section in \onecolumn. If needed, you may revert to \twocolumn when reaching the next page.

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Appendix C: Wide tables and figures after an appendix title: alternate method

To prevent a blank page, a second method is to insert the appendix title after declaring the onecolumn float.
This method should be reserved to appendices containing only one-column floats{figure*} or {table*} and no text.

Table C.1: A one-column {table*}

ISO-L1551	$F_{6.7}$ [mJy]	$\alpha_{6.7-14.3}$	YSO type[d]	Status	Comments
New YSO candidates					
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
Previously known YSOs					
61	0.89 ± 0.58	1.77	Class I	HH 30	Circumstellar disk
96	38.34 ± 0.71	37.5	Class II	MHO 5	Spectral type

Appendix D: Long tables in appendices

For long tables (multipage) in appendices, we use the method described in appendix A. For long landscape tables, please refer to Appendix E.

Table D.1: A long table

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86 [1]	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230

[1] Source not included in the HRI catalog. See Sect. 5.4.2 for details.

Table D.1: continued.

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230

Appendix E: Rotated single page tables

To prevent a blank page with {sidewaystable*}, we use the method described in appendix B: declare the table first, and the section second.

Table E.1: A rotated table with {sidewaystable*}

ISO-L1551	$F_{6.7}$ [mJy]	$\alpha_{6.7-14.3}$	YSO type[d]	Status	Comments
		New YSO candidates			
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
		Previously known YSOs			
61	0.89 ± 0.58	1.77	Class I	HH 30	Circumstellar disk
96	38.34 ± 0.71	37.5	Class II	MHO 5	Spectral type

Appendix F: Rotated long tables in appendices

For rotated long tables in appendices, we use the method described in appendix A, combined with {landscape}.

Table F.1: A long landscape table

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86 [2]	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686

[2] Source not included in the HRI catalog. See Sect. 5.4.2 for details.

Table F.1: continued.

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230